

Semester Two 2026
Mid-semester SOLUTION

Faculty of Business & Economics

UNIT CODES: ETC5521
TITLE OF PAPER: Diving Deeper into Data Exploration
EXAM DURATION: 1 hour 30 minutes

RULES

This is a closed book exam, however you may bring one one-sided page of notes equivalent to 12 point font. All answers are to be written on the exam script, in pen or pencil. All other devices need to be out of reach. You must comply with any instructions given to you by Monash exam staff.

As a student, and under Monash University's Student Academic Integrity procedure, you must undertake all your assessments with honesty and integrity. You must not allow anyone else to do work for you and you must not do any work for others. You must not contact, or attempt to contact, another person in an attempt to gain unfair advantage during your assessment. Assessors may take reasonable steps to check that your work displays the expected standards of academic integrity.

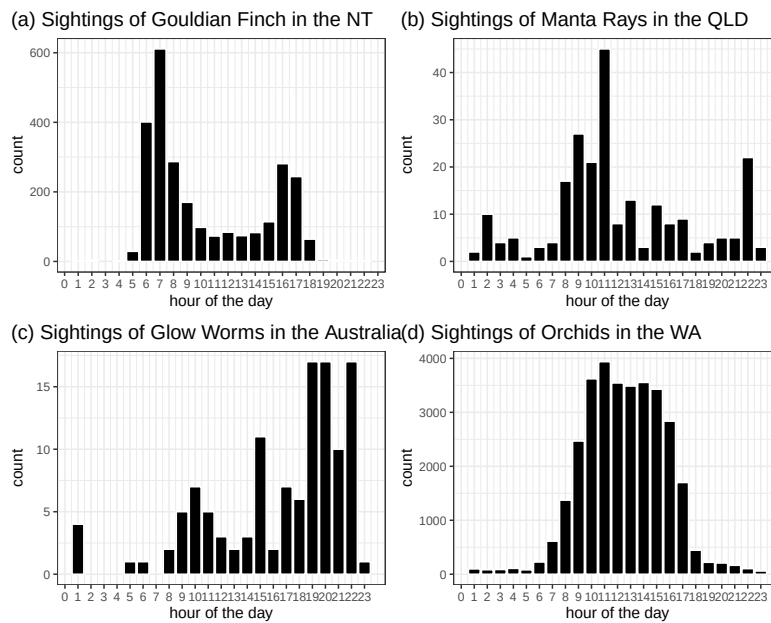
Failure to comply with the above instructions, or attempting to cheat or cheating in an assessment may constitute a breach of instructions under regulation 23 of the Monash University (Academic Board) Regulations or may constitute an act of academic misconduct under Part 7 of the Monash University (Council) Regulations.

There are **11 questions** and a **total of 25 marks**.

Write your Monash ID here:

1. Exploratory data analysis (2pts)

From these four plots, which (if any) shows **an interesting pattern** and why?



Answer: They all have interesting temporal patterns suggesting peak times for viewing.

Plot a shows peaks in viewing Gouldian finches in the early morning and late afternoon.

Plot d shows that peak viewing times are during the day hours 9-5.

Plot c is less reliable as the pattern is more erratic, but suggests peak viewing in the morning hours 8-11am. There's a weird spike at 10pm at night.

Plot d is not so reliable, because the sample size is small. There's a hint of peak viewing in the evening hours 7-10pm, and also a small morning peak around 10am.

2. Exploratory data analysis (2pts)

From the stem-and-leaf compute the lower and upper hinge scores. Use these to determine the length of the box, and compute the inner fence for the highest values of the data. Are any of the high values outside the inner fence? (Don't include the median in the data halves used to compute the hinges.)

The leaf is at one digit, so 10 | 4 means 104.

1 | 2: represents 12

leaf unit: 1

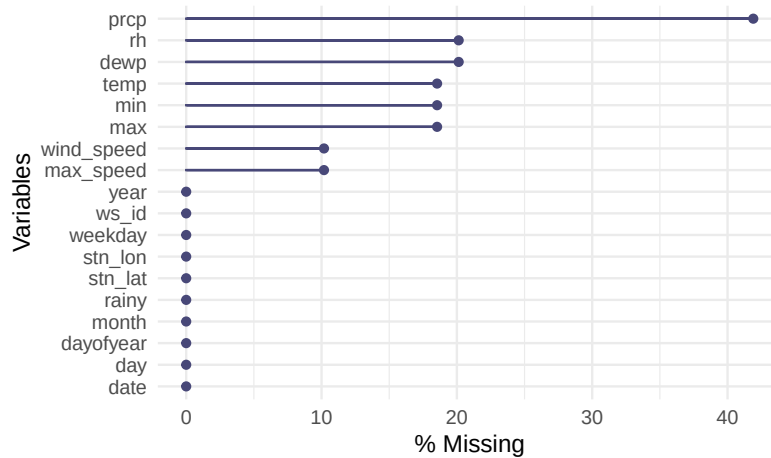
```

      n: 131
40    0 | 1111111111122233445555555566677778888999
(32)  1 | 00000001111222333345566667777899
59    2 | 0000012366788999
43    3 | 0012555688999
30    4 | 1334469
23    5 | 269
20    6 | 38
18    7 | 34799
13    8 | 267
10    9 | 1223
6    10 | 4
      11 |
5    12 | 28
      13 |
3    14 | 8
      15 |
2    16 | 3
      17 |
1    18 | 9
```

Answer: The median is the 66th observation, which is 17 here. The lower hinge is the 33rd, which is 7, and the upper hinge is the 33rd from the top, which is 39. The length of the box is $39 - 7 = 32$. Then the inner fence is $39 + 1.5 * 32 = 87$. There are numerous values above the upper inner fence.

3. Initial data analysis (2pts)

From the following plot, how many observations have missing values? (The number of observations in the data set is 46,390. No need to calculate exactly, approximately is sufficient.)



Answer: There are about 42% on prcp, so at least this many, $0.42 \times 46390 = 19,484$. It may be the same observations that have missings on rh, dewp, temp, min, max, wind_speed, max_speed.

4. Initial data analysis (2pts)

The main pattern here is sightings of manta rays (a marine animal) around Australia. Is there anything unusual? If so, explain.

Sightings of Manta Rays

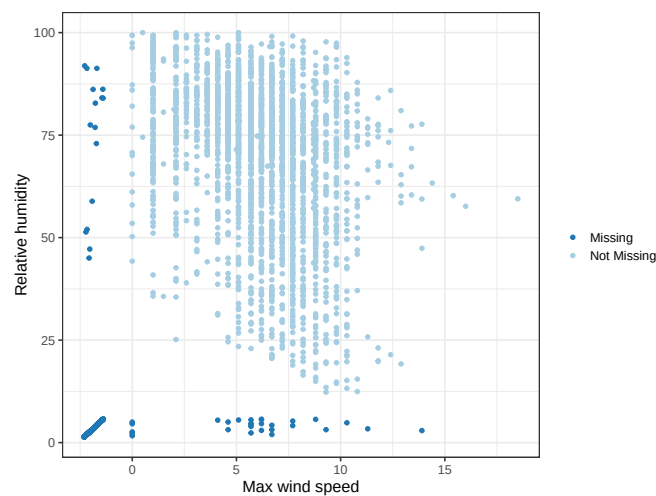


Answer: There is one observation which is clearly not in the ocean, which is implausible. The others are all reasonably likely sightings of a marine animal.

5. Initial data analysis (2pts)

This is a missing value plot of weather variables in Western Australia. What can we say about the missing values in this data? (Circle the ones that apply.)

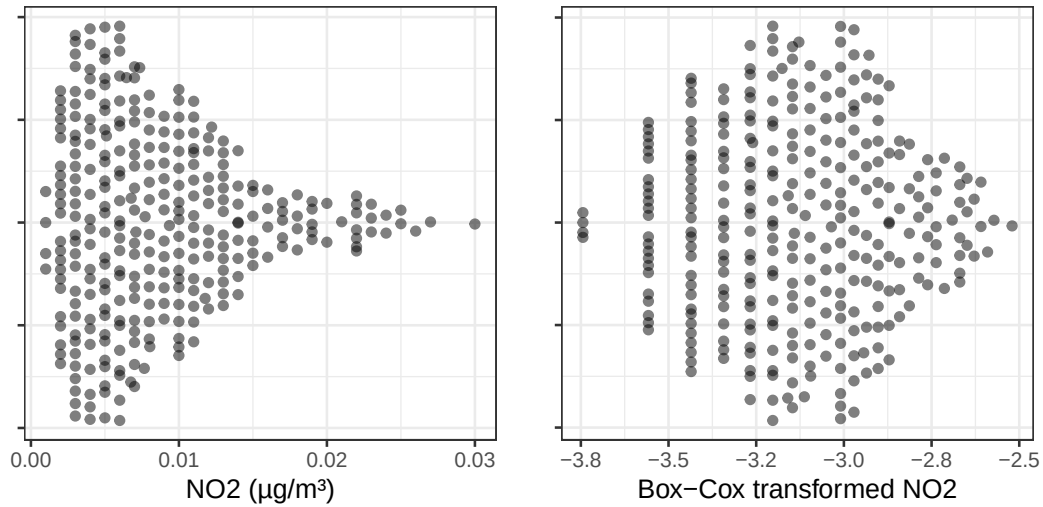
- a. There are missing values on both variables.
- b. There are missing values on only on relative humidity.
- c. There are missing values on only on max wind speed.
- d. The missing values on max wind speed have a similar relative humidity distribution to the complete values.
- e. The missing values on relative humidity have a similar max wind speed distribution to the complete values.
- f. The missing values on relative humidity do not have a similar max wind speed distribution to the complete values.
- g. The missing values on max wind speed do not have a similar relative humidity distribution to the complete values.



Answer: a, d, e are correct. We might argue about d or e, because there are small differences between the complete and missing cases.

6. Initial data analysis (3pts)

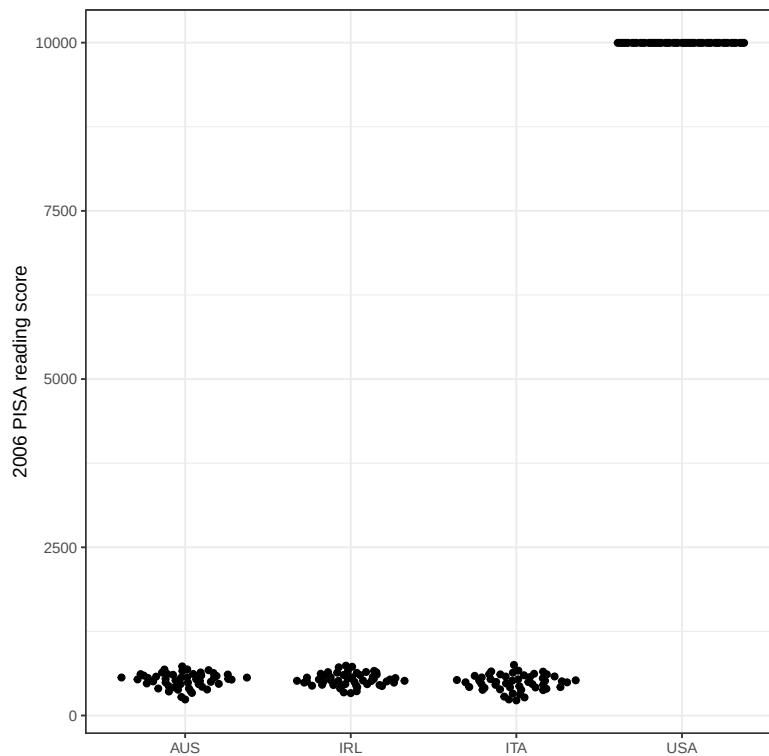
These plots show the data distribution and their distribution after transformation. It uses the ladder of transformations, -3, -2, -1, 0(log), 1/4, 1/3 1/2, 1, 2, 3 but the power function is optimised using Box-Cox. What power value do you think was used for the transformation? And explain anything else that might have been done in the calculation.



Answer: The data was transformed with a value less than 1, probably around 1/4. The data was not shifted before transforming, because all values are above 0.

7. Initial data analysis (3pts)

This is a plot of reading scores from the Program for International Student Assessment from 2006. Describe the pattern in the data, anything unusual, and how you might write a validator to catch any future problems.



Answer: The USA has scores close to 10000, which is implausible. We would suspect that this is an incorrectly handled missing value at 9999. A validator could be used to ensure values range between 0 and 1000, the usual range of test scores.

8. Using computational methods to assess significance of patterns (2pts)

When showing a lineup containing 25 plots to 26 people, 3 people identified the data plot. The p -value is calculated with this code, `pvisual(3, 26, 25)`, producing the following result. Is this evidence to support that the structure in the data plot is different from null structure? (Assume that the nulls were generated correctly.)

```
x simulated binom  
[1,] 3      0.11 0.084
```

Answer: No, because the p -value is not small enough, which suggests its not that rare for 3 out of 26 people to identify the data plot if it was a null sample.

9. Using computational methods to assess significance of patterns (3pts)

Do these two plot definitions test the same null hypothesis about possible structure in the data? Explain your reasoning.

A.

```
ggplot(lineup(null_permute("math"), stu_2022_oz),  
  aes(x=television,  
    y=math)) +  
geom_boxplot(alpha=0.5) +  
facet_wrap(~.sample)
```

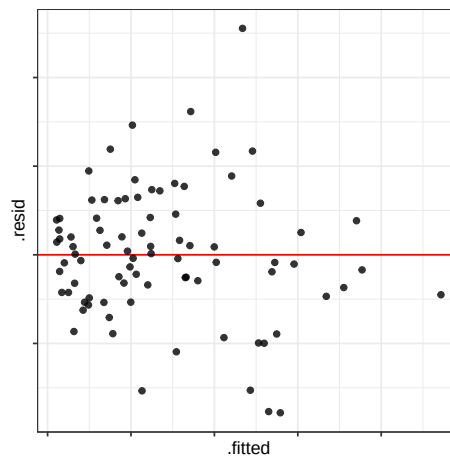
B.

```
ggplot(lineup(null_permute("math"), stu_2022_oz),  
  aes(x=television,  
    y=math)) +  
geom_lv(aes(fill = after_stat(LV)), k=2) +  
facet_wrap(~.sample)
```

Answer: Yes. Both plots show the five number summary (k=2 on the lvplot is the 5 number summary), so they are testing whether this is the same across number of TVs. Points awarded if didn't know what k=2 meant but explanation included the possibility that this was also showing quartiles.

10. Using computational methods to assess significance of patterns (2pts)

The following plot shows residuals plotted against fitted values after a linear model fit. Is there a problem with heteroskedasticity? (Note: Heteroskedasticity means that the variance of the y variable is related to the value of the x variable.)



Answer: Yes. There are a large number of observations where the variance is small, so the difference in variance from left to right is not due to sample size.

11. Using computational methods to assess significance of patterns (2pts)

Here is a snippet of code to make a plot, where v1 is a numeric variable.

```
ggplot(data) +  
  geom_histogram(aes(x=v1))
```

Which of these is the correct null hypothesis, that the plot is testing? And what would be an appropriate method to generate null samples? (Circle your answers for each.)

Possible null hypotheses:

- a. There is a difference between the distribution of v1 across the groups.
- b. There is no difference between the distribution of v1 across the groups.
- c. There is a difference between the means of v1 across the groups.
- d. There is no difference between the means of v1 across the groups.
- e. The distribution of the values in v1 are consistent with a sample from a given statistical distribution.
- f. The distribution of the values in v1 are not consistent with a sample from a given statistical distribution.
- g. The quantiles of v1 are consistent with a sample from a given statistical distribution.
- h. The quantiles of v1 are not consistent with a sample from a given statistical distribution.

Possible null generating methods:

- i. `null_permute()`: Permute the v1 variable
- ii. `null_dist(v1, "exp")`: Generate a sample from an exponential distribution for x1, for the different classes.
- iii. `null_dist(v1, "norm")`: Generate a sample from a normal distribution for v1, for the different classes.
- iv. `null_lm("rotate")`: Generate residuals by fitting v1 to class, using the rotation method.
- v. `null_lm("bootstrap")`: Generate residuals by fitting v1 to class, using the bootstrap method.

Answer: The plot is making a single histogram of the variable v1. This means that the null has to be about a distribution. The null is always stating nothing is interesting, so that the distribution could be described as a sample from a particular distribution would be the not interesting case. So e and ii or iii are the correct answers.

Total: 25
